Discourse coherence modulates use of predictive processing during sentence comprehension

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Background

Contextual cues influence predictive processing, yet there is a lack of consensus as to the divergent roles of global and local contexts in informing expectations of upcoming material.

Coherence across a discourse influences the event-based representation built, and can either signal maintenance or shifting of the current situation model.

This study investigates how global and local contexts interact during predictive processing, and whether coherence is a valid cue that comprehenders use in forming expectations.

Stimuli

Condition	Context Sentences	Target Sentence
High Coherence, High Predictability	Crime was a problem. Residents were starting to feel unsafe.	The police arrested the local gang for selling **drugs** on the corner.
High Coherence, Low Predictability		The police arrested the local gang for selling **toys** on the corner.
Low Coherence, High Predictability	The city was a safe place. It had a low crime rate and high employment.	The police arrested the local gang for selling **drugs** on the corner.
Low Coherence, Low Predictability		The police arrested the local gang for selling **toys** on the corner.

Anomalous: The police arrested the local gang for selling **words** on the corner.

Experiments

Task: online self-paced reading experiment → measured reading times (ms) by keypress

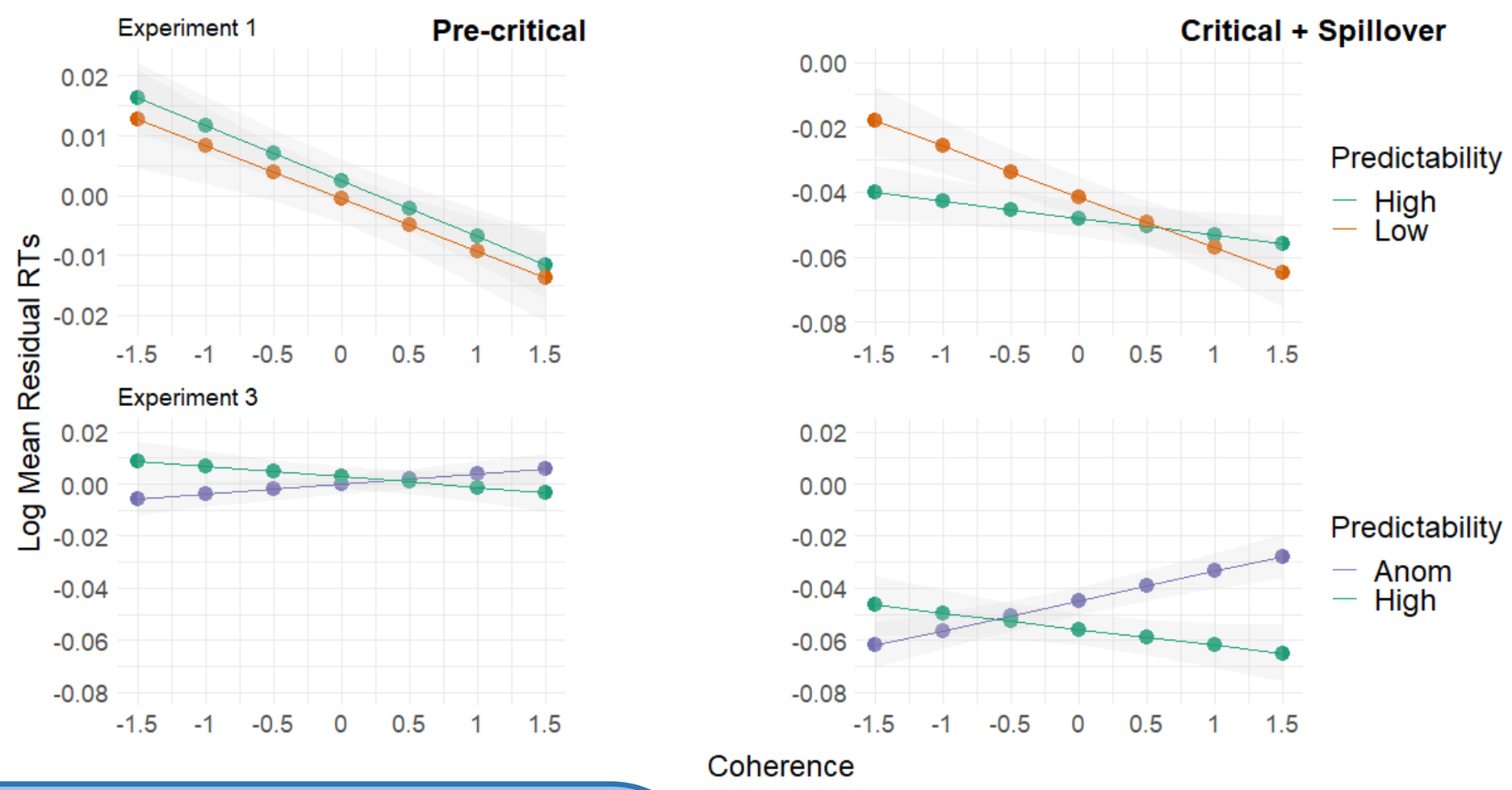
Analysis: linear mixed effects models

Experiment 1: 2x2 factorial design (n=84)

Experiment 2: 75/25 HC/LC contexts (n=108)

Experiment 3: replace LP with Anomalous condition (n=97)

Results



Exp 1

Main effect of Coherence: faster RTs for target sentences preceded by HC

Exp 2

Similar results to Experiment 1

Exp 3

Main effect of Predictability: faster RTs for HP words than Anomalous

Interaction of Coherence and Predictability: effect size depends on coherence; faster for HC/HP, but slowdown for HC/Anomalous

Post-hoc tests: when separated by Coherence, Predictability effect present for more coherent trials, but not less coherent

Discussion

- Comprehenders are sensitive to subtle shifts in the coherence of the wider narrative and will use cues from both levels (global and local), especially when violations present in local context.
- Findings align with current theories on flexible nature of predictive processing.
- Discourse coherence is an informative cue for forming expectations about upcoming material.

Preprint: Github: https://github.com/gacarter/spr OSF: https://osf.io/a8j4c